

RESEARCH PAPER

Deep Learning Approaches for Medical Image Classification: A Comparative Study

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ABSTRACT

This paper presents a comprehensive comparative analysis of deep learning architectures for medical image classification tasks. We evaluate Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and hybrid architectures across three benchmark medical imaging datasets: ChestX-ray14, ISIC 2019 (dermoscopy), and PathMNIST (histopathology). Our experiments demonstrate that hybrid CNN-Transformer models achieve state-of-the-art accuracy of 94.7 percent on chest X-ray classification while maintaining computational efficiency suitable for clinical deployment. We further propose an ensemble strategy combining ResNet-50, EfficientNet-B4, and Swin Transformer that yields a 2.3 percent improvement over individual model baselines. Statistical significance tests confirm the robustness of our findings across all evaluation metrics including AUC-ROC, F1-score, and sensitivity.

Keywords: Deep Learning, Medical Imaging, CNN, Vision Transformer, Transfer Learning, Classification

1. INTRODUCTION

Medical image analysis has witnessed transformative advances through the application of deep learning techniques. Traditional computer-aided diagnosis systems relied heavily on handcrafted features and domain-specific preprocessing pipelines. The emergence of convolutional neural networks revolutionized this paradigm by enabling end-to-end feature learning directly from raw pixel data. Recent developments in attention mechanisms and transformer architectures have further expanded the methodological toolkit available to researchers. This study systematically compares these architectural families across standardized evaluation protocols.

2. RELATED WORK

Prior work in medical image classification includes the seminal CheXNet architecture by Rajpurkar et al. which demonstrated radiologist-level performance on pneumonia detection. Subsequent studies explored transfer learning from ImageNet-pretrained models, with DenseNet-121 and InceptionV3 emerging as popular backbone choices. The introduction of Vision Transformers by Dosovitskiy et al. opened new avenues for capturing global contextual information in medical images. Recent hybrid approaches such as TransMed and MedViT attempt to combine the inductive biases of CNNs with the long-range dependency modeling of transformers.

3. METHODOLOGY

We adopt a standardized experimental framework comprising three phases: data